Cross-species ncRNA annotation using synteny-constrained embedding similarity

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Abstract

Reference-based annotation transfer via orthology has been highly successful for protein-coding genes but remains challenging for non-coding RNAs due to weak sequence conservation, ambiguous transcript boundaries, and incomplete annotations. Here we present CURIA, a pipeline for cross-species ncRNA annotation that combines genome alignment chains with embedding-based representations derived from RNA foundation models. CURIA formulates cross-species ncRNA correspondence as synteny-constrained matching in embedding space. It first identifies candidate loci using a simplified TOGA-based projection, then detects localized subregions ("islands") within transcripts, and finally establishes cross-species correspondence using distribution-level similarity of embedding representations. Using a conservative annotation-based evaluation across ten mammalian genomes, CURIA identified short ncRNA loci, 53.1% of which overlapped existing annotations, and uncovered ~700 candidate conserved intergenic lncRNA loci absent from current annotations. These results suggest that combining synteny with embedding-based similarity can support cross-species ncRNA annotation in settings where primary-sequence conservation alone is insufficient, establishing a proof of concept for integrating comparative genomics with RNA foundation-model embeddings for genome-scale ncRNA analysis.

1. Introduction

1.1 Protein-coding orthology as a tractable problem

Orthology inference for protein-coding genes is well established and supported by scalable annotation transfer pipelines[1–7]. Strong sequence conservation, together with constraints imposed by open reading frames, enables reliable cross-species mapping of gene loci[2,4]. Methods such as TOGA, which combine genome alignment chains with coding sequence structure, achieve high accuracy at genome scale. However, these approaches rely on properties specific to protein-coding genes and do not generalize to non-coding RNAs. In particular, they assume conserved coding structure and detectable sequence similarity, which are often absent outside protein-coding contexts.

In this work, we do not attempt to infer evolutionary orthology. Instead, we focus on identifying candidate cross-species correspondences supported by genomic synteny and consistency in embedding space. This formulation is best viewed as synteny-constrained annotation transfer, where predicted loci represent candidate corresponding elements rather than definitive orthologs.

1.2 The ncRNA orthology challenge

Establishing cross-species correspondence for ncRNAs remains substantially more difficult. Sequence conservation is often weak or absent across evolutionary distances[8–13], and transcript boundaries

are frequently ambiguous, making it unclear what constitutes a conserved unit[8–10,14]. Functional constraints are not reliably captured by primary sequence alone, limiting the applicability of sequence-based approaches[11]. In addition, ncRNA annotations are incomplete and heterogeneous across species[14,15], complicating both evaluation and transfer. Finally, embedding representations of short ncRNAs are context-dependent[16], making direct comparison sensitive to boundary definition.

For example, two ncRNA loci may exhibit low primary sequence similarity while preserving a similar structural pattern:

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GAGGUAUCCAAGAUACCUC
(((((((.....)))))))
AUGCUACGAUCGGUAGCAU
(((((((.....)))))))
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In such cases, sequence-based comparison may fail to establish correspondence, whereas embedding-based representations may capture similarities associated with shared structural patterns.

1.3 Limitations of existing approaches

Existing approaches address complementary aspects of ncRNA detection and comparison but do not provide a scalable framework for genome-wide ncRNA correspondence analysis.

Covariance models (e.g., Infernal[17]/Rfam[18]) offer high specificity but rely on curated families and do not generalize to genome-wide discovery.

Structure-based alignment methods such as LocARNA[19] and RNAforester[20] capture structural similarity but are computationally expensive and not suitable for large-scale analysis. Genome-wide methods such as RNAz[21], EvoFold[22], and FEELnc[23] identify candidate RNA elements or lncRNA loci, but focus on detection or annotation rather than establishing cross-species correspondence. Comparative genome alignments define candidate homologous intervals but do not provide an RNA-specific measure of similarity or a principled way to localize relevant signal[24,25]. As a result, existing approaches either lack scalability, require strong prior knowledge, or do not establish explicit correspondence between loci at genome scale. Together, these limitations motivate a formulation that (i) restricts the search space via synteny and (ii) defines similarity in a representation space not tied to primary sequence.

1.4 Synteny-constrained embedding matching

A key missing component is the integration of synteny with embedding-based similarity. Genome alignment chains define a constrained search space of candidate loci, reducing the combinatorial complexity of cross-species comparison[24,25]. At the same time, RNA foundation models such as RNA-FM[16], RiNALMo[26], and related embedding approaches such as RNABERT[27] provide learned representations associated with RNA family identity, structure-related signal, and downstream predictive tasks[28]. Without synteny constraints, embedding-based matching would require genome-wide comparison over a vastly larger search space, making the problem computationally expensive and increasing the risk of false positives. Combining these sources of information enables tractable comparison of ncRNA loci in settings where primary-sequence conservation is limited.

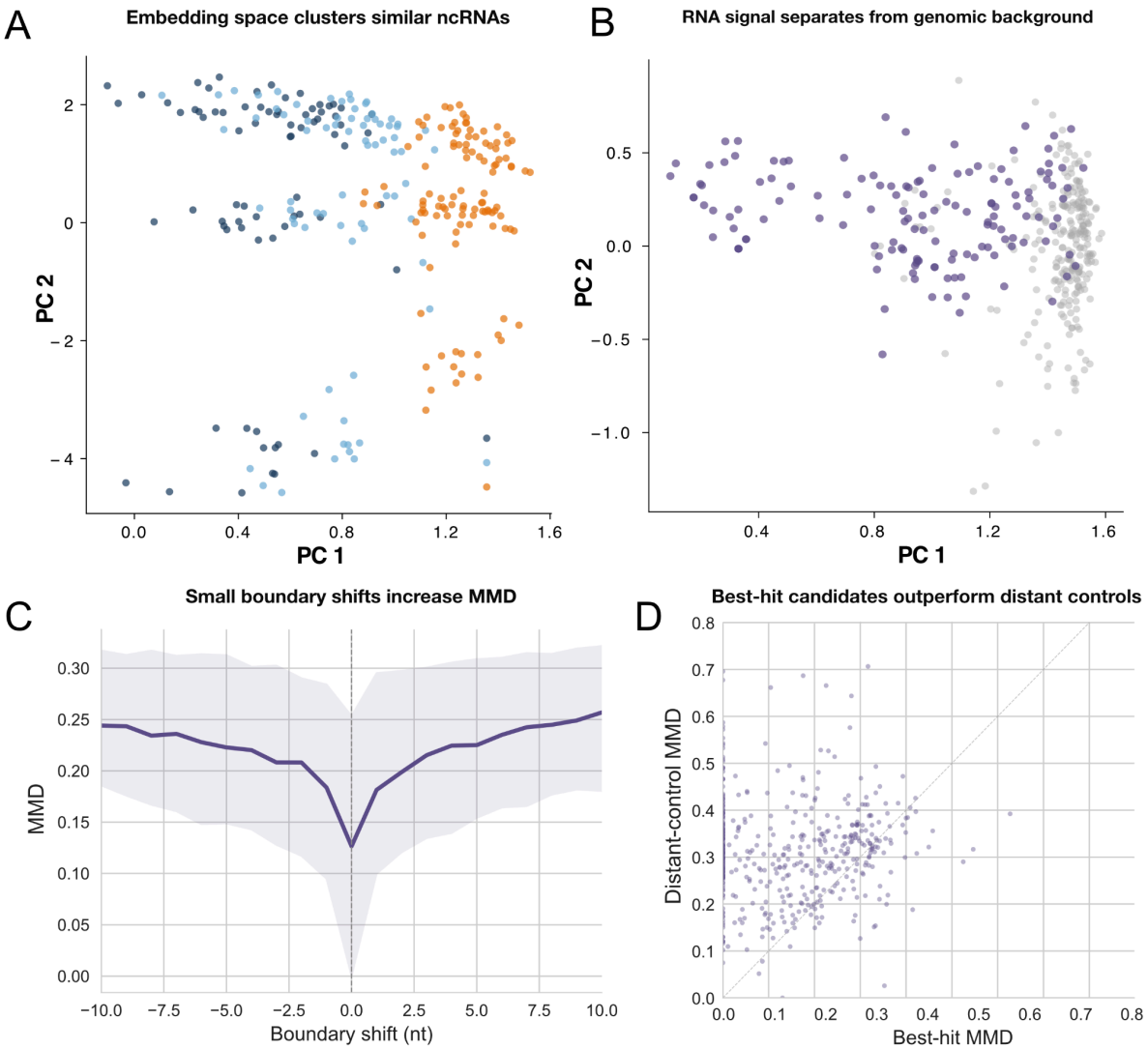

Figure 1. Embedding representations reflect structure-related similarity of ncRNAs and support local matching.
(A) Per-nucleotide RNA-FM embeddings (PCA) for two tRNA copies (blue) and one miRNA (orange). Each point corresponds to a nucleotide. tRNA embeddings cluster together, while the miRNA forms a distinct cluster.
(B) Mean-pooled embeddings for annotated ncRNAs (purple) and background sequences (gray; intergenic and dinucleotide-shuffled). Each point represents one sequence. RNA-derived sequences separate from genomic background despite reduced variance after pooling.
(C) MMD as a function of boundary shift (±10 nt) for predicted loci. Similarity is minimized at the original position and degrades with positional perturbation.
(D) Best-hit MMD compared to a non-overlapping control region (160–320 nt away) within the same locus. Most points lie above the diagonal (85%), indicating that best-supported matches are more similar than nearby alternatives.

1.5 Method overview

We formulate cross-species ncRNA correspondence as matching of embedding-supported regions constrained by genome alignment chains. Candidate loci are first identified using a TOGA-inspired projection that leverages syntenic consistency and coverage of neutrally evolving sequence (e.g., intronic and flanking regions), which remains more reliably alignable across species[1,2].

Within these candidate regions, CURIA evaluates similarity in embedding space rather than relying on primary-sequence alignment. Each region is represented as a set of per-position embedding vectors, and similarity between regions is quantified using Maximum Mean Discrepancy (MMD)[29], a distribution-level measure over embeddings. Low MMD values indicate similar embedding distributions, whereas higher values correspond to dissimilar or background signal. This formulation avoids explicit sequence alignment and provides a continuous similarity score for ranking candidate matches, making it suitable for ncRNAs where primary-sequence conservation is weak.

1.6 Contributions

In this work, we:
(1) Reformulate cross-species ncRNA correspondence as synteny-constrained matching in embedding space.
(2) Introduce CURIA, a pipeline combining chain-based locus projection with embedding-based detection and comparison of localized signal.
(3) Propose an island-based representation for long ncRNAs to localize cross-species signal.
(4) Develop a boundary refinement strategy for short ncRNAs.
(5) Demonstrate a genome-scale proof of concept across multiple mammalian genomes, showing that this approach can recover cross-species ncRNA correspondence in cases where primary-sequence similarity alone is limited.

2. Methods

2.1 Pipeline overview

CURIA is a computational pipeline for genome-scale cross-species ncRNA correspondence that combines chain-guided locus prediction with embedding-based comparison (Figure 2). Given a reference genome with ncRNA annotations and one or more query genomes linked by genome alignment chains, CURIA produces: (i) predicted corresponding loci, (ii) an embedding-based similarity score (MMD), and, for long ncRNAs, (iii) matched localized subregions (“islands”).
The pipeline follows a branching architecture after an initial projection step, handling short and long ncRNAs separately. Short ncRNAs yield a single refined locus, whereas long ncRNAs yield a set of island-level alignments. The pipeline consists of seven stages:
(1) Preprocessing. Overlapping isoforms are collapsed into a union transcript per locus.
(2) Candidate region prediction. Genome alignment chains are used to identify candidate query regions via a simplified TOGA-inspired classifier (Section 2.3).
(3) Short ncRNA refinement. Candidate loci are refined using embedding-based local boundary search (Section 2.4).
(4) Reference island detection. Localized subregions (“islands”) are detected within transcripts (Section 2.5).
(5) Chain-guided projection. Reference islands are projected to the query genome to define targeted search regions (Section 2.6).
(6) Query island detection. Candidate islands are identified in projected regions (Section 2.7).
(7) Island alignment. Reference and query islands are matched using MMD-based comparison (Section 2.8).

Outputs include candidate region assignments, refined short ncRNA loci, and island-level alignments.

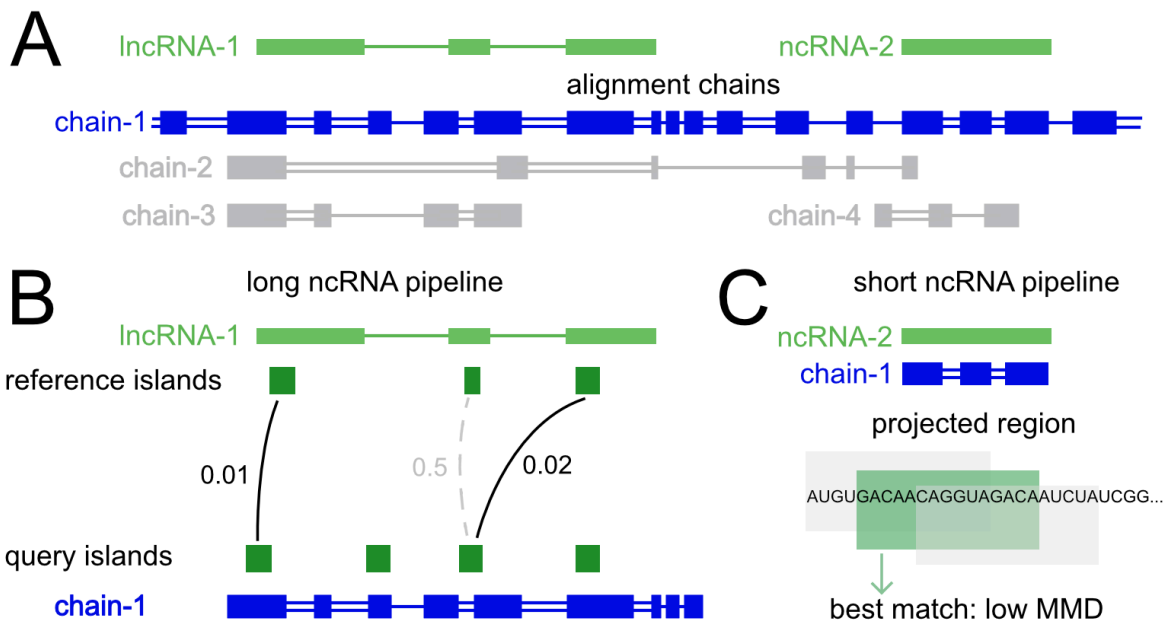

Figure 2. Overview of the CURIA pipeline for cross-species ncRNA correspondence analysis.
(A) Genome alignment chains restrict the search space by defining candidate loci between reference and query genomes.
(B) For long ncRNAs, localized subregions ("islands") are detected and matched across species using embedding-based similarity, enabling localization of candidate conserved elements within transcripts.
(C) For short ncRNAs, correspondence is resolved via local boundary refinement using sliding-window matching in embedding space, selecting the best-supported query region based on MMD.

2.2 Input data

CURIA requires three types of input data:
(1) Reference genome and annotation. The reference genome is provided in 2bit format, with ncRNA annotations in BED12 format including transcript_id, gene_id, and biotype. Protein-coding transcripts are used during candidate region prediction to compute synteny-related features, but are excluded from downstream analysis.
(2) Query genomes. Query genomes are provided in 2bit format. Query-side annotations are not required for prediction but may be used for evaluation.
(3) Genome alignment chains. Pairwise genome alignment chains (UCSC chain format) define syntenic relationships between reference and query genomes. Low-scoring chains are filtered during loading.

2.3 Candidate region prediction

Input. Union transcripts and genome alignment chains.
Operation. CURIA uses a simplified TOGA-inspired chain-guided projection step[1] to define coarse candidate regions for ncRNA loci rather than full transcript reconstructions (Figure 2A).
For each chain–transcript pair, a small set of features is computed:
(1) synteny, defined as the number of distinct genes intersecting the same alignment chain (log-transformed);
(2) global exonic occupancy, defined as the fraction of aligned chain bases overlapping annotated exons;

(3) flank coverage, defined as the fraction of ±10 kb flanking regions covered by chain alignment blocks.
A logistic regression classifier assigns a correspondence probability based on these features. The classifier was trained on labeled chain–transcript pairs derived from human–mouse projections (410,776 pairs; 79,065 orthologous and 331,711 paralogous). On a held-out test set, the classifier achieved ROC-AUC = 0.982. Pairs with predicted probability ≥ 0.5 were retained.
For each retained pair, candidate query intervals were obtained by projecting the transcript through the corresponding alignment chain. Intervals were defined as the span of chain blocks overlapping the transcript, including flanking regions to account for alignment imprecision. When multiple chains satisfied the threshold for a given transcript, all corresponding intervals were retained.
Output. Projected candidate query regions with associated metadata.
Rationale. Only coarse locus-level projections are required to constrain downstream analysis, avoiding assumptions related to coding sequence structure and transcript reconstruction.

2.4 Short ncRNA pipeline

Short ncRNAs (e.g., miRNAs, tRNAs, snoRNAs, snRNAs) typically occupy compact loci, allowing direct local comparison without island-based decomposition (Figure 2C). This branch processes single-exon ncRNAs up to 160 nt. For each locus, the reference sequence is embedded using RNA-FM. The corresponding query region is extended symmetrically to account for alignment imprecision. Because embedding representations are context-dependent[16,27], CURIA performs local boundary search around each projected candidate locus. Candidate windows matching the reference length are tiled across the query region and compared to the reference using MMD (Section 2.8). The best match is refined by local boundary perturbation. Embeddings are projected from 640 to 16 dimensions using PCA prior to comparison (Section 2.10).
Output. Refined query coordinates with an MMD-based similarity score.

2.5 Reference lncRNA island detection

To detect RNA-like signal, we trained a logistic regression classifier on fixed-length sequence windows represented in RNA-FM embedding space. Positive examples consisted of ncRNA-derived windows (including lncRNAs, miRNAs, snoRNAs, snRNAs, scaRNAs, and misc_RNAs), while negative examples consisted of intergenic, non-exonic genomic windows. All sequences were clipped to 72 nt, embedded with RNA-FM, mean-pooled across positions, and projected from 640 to 16 dimensions using PCA prior to training. The classifier was trained on a dataset of 9,620 windows (3,696 signal / 5,924 noise), split 80/20 into training and test sets. On the held-out test set, the classifier achieved ROC-AUC = 0.983.
The resulting model distinguishes embedding patterns associated with annotated ncRNAs from genomic background and is used as a lightweight detector of localized RNA-like signal rather than a general ncRNA annotation model[19,20].
Each reference union transcript is scanned using a sliding window (72 nt, stride 16 nt). Windows are embedded, projected to PCA space, and scored by the classifier. The resulting probability trace is smoothed using a moving average filter (window size = 5), and contiguous regions exceeding a fixed probability threshold (0.25) are defined as candidate islands.
Island coordinates are mapped to genomic space, allowing multi-segment islands across splice junctions.
Output. Reference islands with genomic coordinates and associated scores.

2.6 Chain-guided projection of islands

Input. Reference islands, genome alignment chains, and candidate region assignments.
Reference islands are projected through alignment chains to define candidate query regions (Figure 2A). Intervals are extended to account for alignment imprecision. Projected intervals on the same chromosome and strand are merged into non-redundant regions while retaining links to the originating reference islands.
Output. Query regions with mappings to reference islands.
Rationale. Projection restricts downstream search to compact regions while preserving localized embedding-supported signal.

2.7 Local query island evaluation

Input. Merged query regions, query genome, and the classifier from Section 2.5.
Each query region is scanned using the same procedure as reference island detection. Windows are embedded, projected, and scored, and contiguous high-scoring regions are defined as islands.
Basic filtering removes regions with implausible size or lacking corresponding reference signal.
Output. Query islands with genomic coordinates and scores.

2.8 Island alignment in embedding space

Window similarity is quantified using Maximum Mean Discrepancy (MMD) with an RBF kernel, treating embeddings as empirical distributions:

$$MMD^2(X,Y) = \frac{1}{n(n-1)}\sum_{i\neq j} k(x_i,x_j) + \frac{1}{m(m-1)}\sum_{i\neq j} k(y_i,y_j) - \frac{2}{nm}\sum_{i,j} k(x_i,y_j)$$

where $k(x, y) = \exp(-\gamma \|x - y\|^2)$.
Here, $X = \{x_1, ..., x_n\}$ and $Y = \{y_1, ..., y_m\}$ denote the sets of per-position embedding vectors for two sequence windows. The kernel bandwidth γ is set using the median pairwise distance heuristic computed on embedding vectors within each comparison.
Reference and query islands are decomposed into fixed-length windows (96 nt, stride 4 nt). For each pair of windows, MMD is computed, yielding an alignment matrix of size (# reference windows × # query windows).
To extract local correspondences, the MMD matrix is converted into a similarity matrix using a linear transformation $S = \tau - MMD$, where τ is a constant offset (empirically set to 0.15). Local alignments are then extracted using a Smith-Waterman procedure[30] with affine gap penalties (gap open = 0.03, gap extend = 0.01).
Candidate matches are filtered using the following constraints: (i) minimum alignment length ≥ 40 nt; (ii) mean MMD ≤ 0.15; and (iii) collinearity, enforced by retaining the largest non-crossing subset of matches via longest increasing (same strand) or longest decreasing (opposite strand) subsequence selection. To avoid redundant matches, each query island is assigned to at most one reference island using a greedy selection procedure that prioritizes lowest mean MMD.
MMD values are reported as empirical distance scores for downstream ranking and filtering of candidate matches.

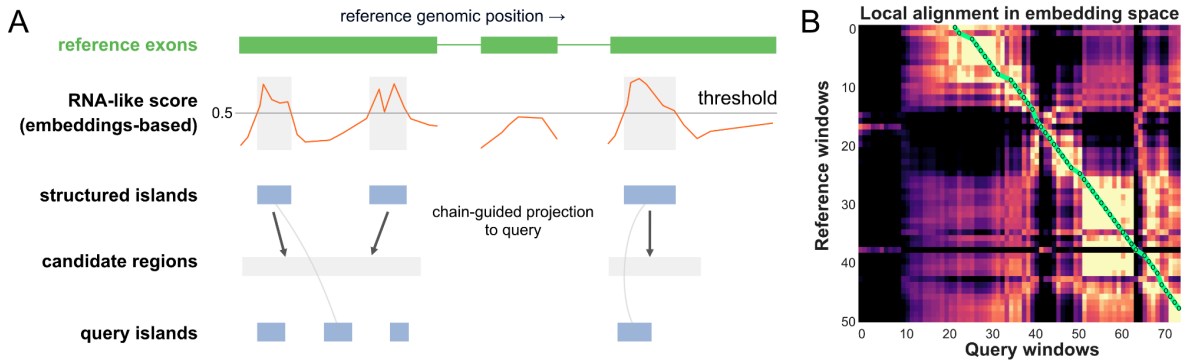

Figure 3. Identification and matching of localized ncRNA subregions using embedding-based similarity.
(A) RNA-like signal along a reference transcript is detected using an embedding-based score and thresholded to define localized subregions ("islands"). These islands are projected via genome alignment chains to define candidate regions in the query genome, where corresponding subregions are identified.
(B) Similarity between reference and query regions is computed as an MMD-based alignment matrix, and local matches are extracted using a Smith–Waterman procedure.

2.9 Evaluation metrics

Evaluation is based on agreement with existing annotations and serves as an approximate proxy for ground truth. Due to incomplete and heterogeneous ncRNA annotation across species, these metrics should be interpreted as measures of annotation agreement rather than direct estimates of biological accuracy.

2.9.1 Short ncRNA evaluation

For short ncRNAs, evaluation is performed at the locus level using overlap with annotated transcripts. A predicted region is considered a match if it overlaps an annotated ncRNA with ≥50% reciprocal overlap. We report:

- annotation agreement: fraction of predictions overlapping annotated loci
- locus recovery: fraction of annotated loci recovered by predictions
- near-complete matches: fraction of predictions with ≥99% overlap

These measures assess how well predicted loci align with annotated boundaries, while accounting for minor positional variation.

2.9.2 lncRNA locus-level evaluation

For lncRNAs, evaluation is performed at the locus level with relaxed criteria due to boundary ambiguity. A predicted locus is considered supported if it overlaps any annotated exon of an ncRNA gene.
We report:

- fraction of loci overlapping annotated exons
- distribution of overlap proportions across predictions

This reflects coarse agreement with annotated loci rather than precise boundary recovery.

2.9.3 Island-level evaluation

For long ncRNAs, evaluation is extended to localized subregions ("islands"). A matched island is considered supported if it overlaps any annotated ncRNA exon in the query genome.
We report:

- fraction of matched islands overlapping annotated regions
- number of matched islands per locus
- cross-species consistency of island matches

This analysis captures cross-species correspondence at the subregion level, independent of full-transcript recovery.

2.9.4 Annotation agreement vs confidence

To assess the relationship between embedding-based similarity and annotation support, we analyze annotation agreement as a function of MMD.
Predictions are stratified by MMD values, and we report:

- fraction of predictions overlapping annotations across MMD bins
- distribution of MMD values for overlapping and non-overlapping predictions

This provides an empirical characterization of MMD as a measure of correspondence confidence, rather than relying on predefined thresholds.

2.9.5 Limitations

Predictions outside annotated regions may represent either false positives or unannotated ncRNAs[10,14,15]. Annotation completeness varies across species, and ncRNA boundaries are often imprecise. Accordingly, reported performance should be interpreted as a lower bound.

2.10 Implementation details

Embeddings are projected from 640 to 16 dimensions using PCA. The PCA transformation was precomputed on a large corpus of RNA-FM embeddings (~10^6 sequence positions sampled across diverse human transcripts) to provide a stable, dataset-independent representation.
Embedding inference is performed on GPU using batched execution, whereas downstream MMD computation, alignment, and filtering are performed on CPU. Internally, CURIA is organized as a staged task pipeline that separates embedding inference from downstream analysis. Sequence windows are processed by a shared GPU worker using micro-batching, while subsequent matching and filtering steps operate on persisted intermediate results. This design improves hardware utilization, supports incremental execution, and allows reference-side preprocessing to be reused across query genomes.
Outputs are reported in standard tabular and BED formats. The use of fixed embedding models and PCA projections ensures consistent representations across runs.

3. Results

3.1 Benchmark setup

CURIA was evaluated on a panel of mammalian genomes, using human (hg38) as the reference and multiple genomes as queries (rhesus macaque (rheMac10), mouse (mm39), rat (rn7), cow (bosTau9), pig (susScr11), horse (equCab3), cat (felCat9), hedgehog (eriEur2), armadillo (dasNov3), and

opossum (monDom5)). Genome assemblies were obtained from NCBI[39] and UCSC Genome Browser[40]. Due to variability in annotation quality across species, high-quality ncRNA annotations were used only for human (GENCODE V48) and mouse (GENCODE VM37)[31], while other species were analyzed without relying on their native annotations.
Human and mouse contain on the order of ~20,000 annotated ncRNA transcripts, spanning both well-supported elements (e.g., tRNAs, rRNAs, selected snoRNAs) and less-characterized transcripts, particularly among lncRNAs.
Evaluation was performed at both the locus level (for short ncRNAs and lncRNA loci) and the substructure level (for island matches), following the criteria defined in Section 2.9.

3.2 Overall performance

We first evaluated agreement between predicted loci and existing annotations. Among 2,805 predicted short ncRNA loci, 53.1% overlap annotated exons, with 47.7% covering ≥50% of the annotated transcript and 35.5% showing near-complete overlap (≥99%). Among overlapping predictions, the median overlap is 100% (mean 88.8%), indicating that matched loci are typically well localized and consistent with annotated boundaries.
These results suggest that matched loci are generally well localized when annotation support is present. The substantial fraction of predictions without annotation overlap is consistent with a combination of annotation incompleteness, projection errors, and potential false positives. Notably, many non-overlapping predictions retain strong syntenic support, suggesting that some may correspond to unannotated or weakly annotated ncRNA loci, although projection errors and false positives are also expected to contribute.

3.3 Short ncRNA performance and MMD behavior

We evaluated similarity between matched short ncRNA loci using MMD in embedding space. Across all predictions, the MMD distribution is skewed toward low values (median 0.108; Fig. 4A), with a substantial fraction of near-zero values, indicating consistent embedding representations between matched reference and query loci.
Stratification by biotype reveals clear differences (Fig. 4C). snoRNAs and miRNAs exhibit the lowest MMD values, consistent with their compact size and well-characterized structural constraints. In contrast, snRNAs and misc_RNAs show higher and broader MMD distributions likely reflecting increased ambiguity due to copy number, sequence divergence, and genomic context. tRNAs serve as a positive control, with consistently low MMD values in line with their well-characterized structural conservation[32].
MMD is strongly associated with annotation agreement. Predictions with low MMD values (≈ <0.1) show high overlap with annotated loci (~90%), whereas higher MMD values show reduced overlap and increased dispersion. Predictions with MMD >0.2 rarely overlap annotations and are enriched for ambiguous cases. These ranges are empirical and used for interpretation rather than as fixed thresholds.
Consistent with this, loci overlapping annotations are enriched at low MMD values, while non-overlapping predictions are shifted toward higher values (Fig. 4B), supporting the interpretation of MMD as a continuous empirical measure of correspondence confidence.
Notably, a subset of high-confidence predictions (MMD ≈ 0) does not overlap existing annotations despite strong syntenic support, suggesting that annotation incompleteness contributes to the observed gap, although projection errors and false positives also contribute.

To assess whether embedding-based similarity merely recapitulates sequence conservation, we compared MMD to pairwise sequence identity for matched short ncRNA loci. The two measures were strongly anticorrelated (Pearson r = -0.797; Spearman ρ = -0.846; n = 2,697), indicating that low MMD generally coincides with higher primary-sequence similarity.

However, this relationship is not deterministic. At intermediate sequence identity levels (≈45–60%), MMD values span a broad range, indicating that loci with comparable sequence similarity can exhibit substantially different embedding-based similarity. This variability suggests that embedding representations capture additional structure beyond raw sequence identity in this regime, providing further discrimination between candidate matches. At higher sequence identity, MMD values converge toward low values, while at lower identity they become more uniformly high, consistent with background signal.

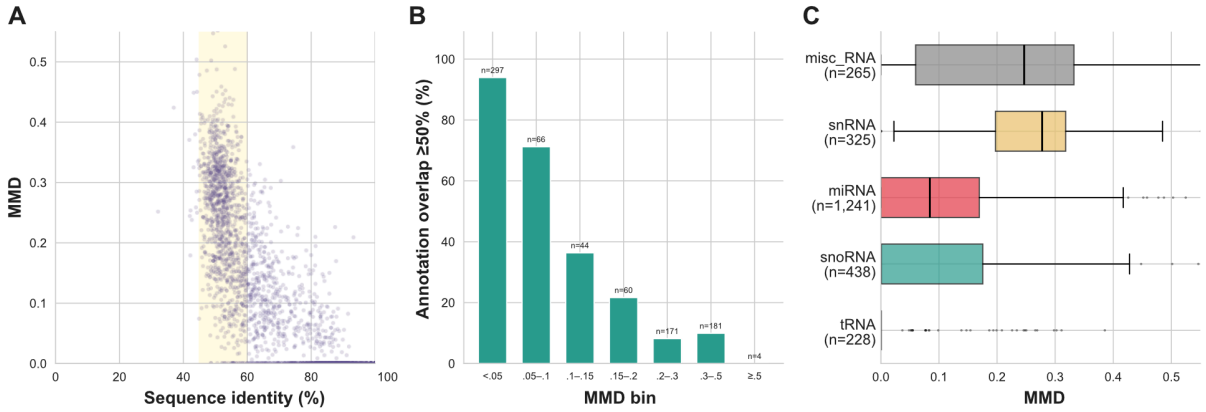

Figure 4. Behavior of embedding-based similarity (MMD) across ncRNA loci.
(A) Relationship between sequence identity and MMD for matched loci. While the two measures are strongly anticorrelated, substantial variability is observed at intermediate sequence identity (~45–60%), indicating that embedding-based similarity is not fully reducible to pairwise primary-sequence identity.
(B) Annotation agreement as a function of MMD. The fraction of loci with ≥50% overlap with annotated ncRNAs decreases with increasing MMD, supporting its interpretation as a continuous empirical measure of correspondence confidence.
(C) Distribution of MMD values across ncRNA biotypes. Compact RNAs (miRNAs, snoRNAs, tRNAs) exhibit low and narrow distributions, whereas snRNAs and other RNA classes show broader and higher MMD values.

3.4 Structural conservation analysis ("islands")

We analyzed localized subregions ("islands") within lncRNAs to assess cross-species correspondence beyond the locus level. Island detection reveals that embedding-supported signal is sparse and localized. Most lncRNAs contain a small number of discrete islands rather than uniform signal across the full transcript, as illustrated by representative loci (Fig. 6A), where embedding-supported signal appears as discrete peaks rather than continuous transcript-wide patterns, indicating that the strongest cross-species signal is concentrated in compact subregions.
Across loci, islands were detected in approximately 74% of lncRNAs, with a median of 2 islands per transcript. The distribution is skewed, with most transcripts containing one or two islands and a minority exhibiting more complex patterns.
To assess cross-species correspondence, we evaluated matches between reference and query islands. At the gene level, 83.9% of lncRNAs exhibit at least one cross-species match, indicating that detectable correspondence is widespread but rarely spans entire transcripts. Correspondence at the

level of these subregions is typically partial, supporting a model in which conserved signal is localized rather than transcript-wide.
Only 22% of matched islands overlap annotated ncRNA regions in the query genome, and 17% show ≥50% exonic overlap, suggesting that many candidate elements are not captured by current annotation resources.
Together, these observations support a modular pattern of cross-species correspondence, with signal concentrated in discrete localized regions rather than uniformly distributed across entire transcripts[10–12,14,33]. This pattern is also reflected in the spatial distribution of cores along transcripts (Fig. 6A), where conserved signal is confined to specific regions rather than spanning full transcript length. This interpretation is broadly consistent with observations from well-characterized lncRNAs such as NEAT1, XIST, and MALAT1, in which localized domains have been linked to distinct architectural, spreading, or stability-related functions[34–36].

3.4.1 Multi-species conserved cores

Here, "conserved cores" refers operationally to islands showing recurrent low-MMD matches across multiple species within syntenically projected loci.
To identify recurrent cross-species elements, we examined islands that are consistently recovered across multiple species. We observe a subset of islands that exhibit reproducible matches across species with low MMD values (Fig. 6B–C), where cores show consistent cross-species similarity and increasing divergence with phylogenetic distance.
Across all detected cores, 83% are recovered in at least two species, 49% in five or more species, and approximately 1% are consistently detected across all ten species, including the marsupial outgroup (opossum). This distribution defines a spectrum of correspondence ranging from weakly shared to highly stable elements (Fig. 6B).
In contrast to locus-level overlap, which is often incomplete or ambiguous, these cores exhibit stable cross-species correspondence at the level of localized embedding similarity. They are frequently detected even in loci where full-transcript correspondence is weak or absent, reinforcing a modular pattern of cross-species correspondence in lncRNAs.
To assess whether such matches could arise by chance, we compared results to a randomized baseline in which query islands were replaced with randomly sampled regions within the same syntenic intervals. Under this null model, consistent low-MMD matches were rare, indicating that recurrent cores are unlikely to arise from random similarity within syntenic regions and instead reflect reproducible embedding-level correspondence.

3.5 Candidate conserved regions absent from current annotations

We next assessed how frequently such candidate elements occur across loci. After excluding regions overlapping or proximal to protein-coding genes, we identify approximately 700 candidate conserved intergenic lncRNA loci exhibiting consistent low-MMD similarity in embedding space across ≥9 species. The full set of loci is available in Data and Code Availability (preprint_results/lncRNAs_with_conserved_cores.txt). These loci represent candidate elements and may include false positives despite filtering.
MMD exhibits an empirical separation regime, with the majority of high-confidence matches falling below ~0.1, beyond which few additional loci are recovered. This suggests a practical regime for identifying reproducible signal in embedding space. The persistence of these loci after filtering suggests that the observed signal is unlikely to be primarily driven by coding-associated conservation, although residual confounding cannot be excluded.

Together, these results suggest that reproducible low-MMD candidate elements extend beyond currently annotated ncRNA catalogs, but require further validation.

3.6 Case studies

To illustrate the behavior of CURIA in representative scenarios, we examined individual loci, including (i) well-supported correspondences with existing annotations, (ii) cases with incomplete or ambiguous annotation, and (iii) loci with structural variation (Fig. 5A–C).

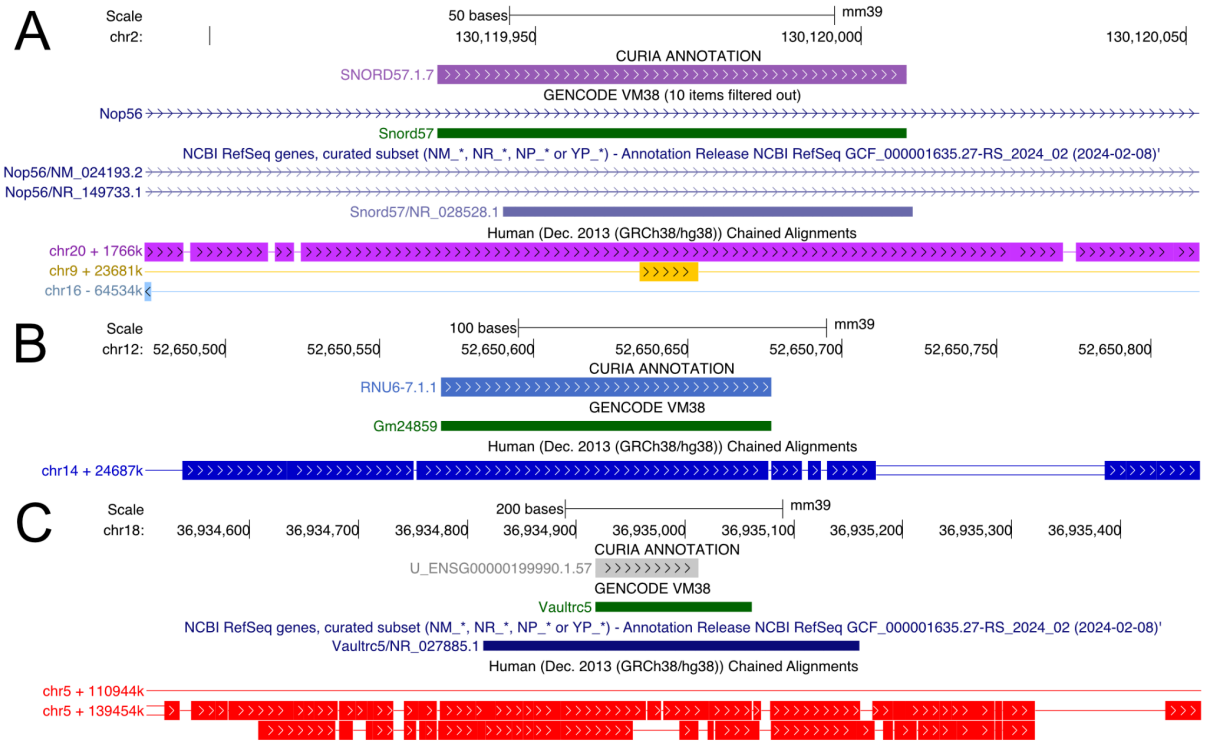

Figure 5. Recovery of short ncRNA loci across species using CURIA.
Genome browser snapshots (UCSC) illustrating representative examples of short ncRNA projection from human (hg38) to mouse (mm39). Tracks show CURIA predictions, GENCODE annotations, and alignment chains.
(A) Projection of human SNORD57 to mouse identifies a nearly exact match overlapping annotated Snord57, demonstrating precise localization under strong structural conservation. (MMD = 0.0)
(B) Projection of human RNU6-7 intersects an annotated mouse locus (Gm24859), indicating recovery of a corresponding genomic region despite ambiguous or incomplete annotation. (MMD = 0.0)
(C) Projection of human vault RNA (ENSG00000199990) identifies a corresponding locus overlapping mouse Vaultrc5. The matched region differs in length from the human transcript, yet CURIA localizes the most similar segment within the syntenic region. (MMD = 0.059)

3.6.1 Clean reference-supported match

We first consider a well-characterized ncRNA locus with consistent annotation across species (Fig. 5A). CURIA recovers a precise correspondence between human *SNORD57* and mouse *Snord57*, with accurate localization and full agreement with existing annotations. This example demonstrates reliable recovery of a well-supported cross-species correspondence under strong structural constraint.

3.6.2 Recovery under incomplete or ambiguous annotation

In some cases, CURIA identifies loci that overlap weakly annotated or ambiguous regions (Fig. 5B). For example, projection of human *RNU6-7* recovers a corresponding mouse locus (*Gm24859*), despite

limited annotation support. This illustrates that CURIA can recover biologically consistent loci even when annotation is incomplete.

3.6.3 Length divergence and partial correspondence

Finally, we consider a case where transcript structure differs between species (Fig. 5C). Projection of human vault RNA identifies a corresponding region overlapping mouse *Vaultrc5*, despite differences in transcript length. CURIA localizes the most similar segment within the syntenic region, indicating that correspondence is driven by localized similarity in embedding space rather than full-length transcript conservation.

3.7 Error analysis

We analyzed failure cases to identify the primary sources of error in CURIA predictions. The dominant failure modes arise from limitations in projection, annotation, and signal strength.
Chain-based projection defines the search space and acts as a hard constraint: if the corresponding region is not captured by genome alignments, it cannot be recovered. In practice, fragmented alignments and duplicated or paralogous regions frequently lead to missing or ambiguous projections.
Incomplete and imprecise reference annotations further limit recoverable signal, particularly for lncRNAs. Missing or truncated annotations often result in partial matches rather than full-length correspondence.
A substantial fraction of loci exhibit weak or diffuse embedding signal. Such cases produce no detectable islands or only fragmented matches, reducing confidence in correspondence. In addition, approximate projection and context-dependent embeddings can affect boundary localization, especially for short ncRNAs.
Overall, most observed failures are associated with missing or ambiguous signal and projection constraints, rather than instability of the downstream matching procedure.

3.8 Runtime and scalability

We evaluated CURIA on a representative genome pair (hg38 vs mm39). Using an AMD EPYC 9575F CPU and an NVIDIA RTX PRO 6000 GPU, the full pipeline completes in ~30 minutes when restricted to primary isoforms, and remains within ~1 hour for a more comprehensive transcript set.
Runtime is dominated by island matching, while upstream steps, including candidate-region prediction and short ncRNA refinement, contribute a smaller fraction of total runtime. Both reference and query island detection are restricted to syntenically projected regions, substantially reducing the effective search space.
Runtime scales primarily with the total exonic sequence in the reference annotation rather than genome size. The number of islands per locus is relatively stable, and no combinatorial explosion is observed, although loci with complex island structure may incur higher cost.
Embedding inference is performed on GPU with batched execution (up to 4096 sequences per batch), while MMD computation and alignment are CPU-bound and represent the primary bottleneck.
Individual island matching jobs are fast (<1 s) but dominate runtime in aggregate.
Overall, CURIA scales approximately linearly with annotation size and can be applied to multi-species panels without prohibitive computational cost on a single modern GPU. This efficiency is enabled by decoupling GPU-bound embedding inference from CPU-bound matching and by restricting computation to syntenically projected regions.

4. Discussion

CURIA should be viewed primarily as a proof of concept demonstrating that synteny-constrained embedding comparison yields interpretable cross-species signal for ncRNA correspondence at genome scale, rather than as a finished annotation framework optimized for exhaustive sensitivity or production use. Within this framing, several key observations emerge.

4.1 Key findings

We show that cross-species ncRNA correspondence can be recovered at genome scale using a combination of syntenic constraints and embedding-based similarity. This enables identification of candidate corresponding loci even in the absence of strong sequence conservation.
For short ncRNAs, correspondence is typically well localized when supported by annotation, and embedding-based similarity (MMD) provides a consistent empirical measure of match confidence. For long ncRNAs, correspondence is frequently partial and localized, with signal concentrated in discrete subregions rather than spanning full transcripts.
Together, these results support a model in which cross-species ncRNA correspondence is heterogeneous and modular, and can be captured using localized comparison in embedding space.

4.2 Comparison to existing methods

Existing approaches provide complementary capabilities for ncRNA detection and structural comparison but do not directly address genome-scale correspondence.
Covariance models and structure alignment methods offer high specificity but rely on curated families or are computationally expensive, limiting their scalability. Genome-wide scanning approaches identify candidate RNA elements but do not establish explicit cross-species correspondence.
In contrast, CURIA combines syntenic constraints with embedding-based similarity to enable tractable genome-scale analysis of ncRNA correspondence without requiring curated families or explicit structural alignment.

4.3 Biological implications

Our results indicate that cross-species ncRNA correspondence is often heterogeneous and localized rather than transcript-wide. For long ncRNAs, correspondence frequently concentrates in compact subregions, while the surrounding transcript context shows weak or no detectable similarity.
Embedding-based similarity (MMD) is strongly associated with sequence identity for short ncRNAs, but this relationship is not uniform. At intermediate sequence identity levels (≈50–60%), substantial variability in MMD is observed, indicating that loci with comparable sequence similarity can exhibit markedly different embedding-based similarity.
Only subsets of transcripts exhibit consistent cross-species correspondence, while other regions show weak or no detectable similarity. This behavior supports a modular pattern, in which localized regions show stronger and more reproducible signal than the surrounding transcript context[10,12,14].
The island-based representation provides a practical way to capture this behavior: recurrent low-MMD islands define candidate conserved cores, while intervening regions appear less constrained.
Correspondence patterns also vary across RNA classes: compact RNAs such as tRNAs and snoRNAs show strong and consistent signal, whereas long ncRNAs more often exhibit partial or fragmented correspondence.

Together, these observations support a model in which candidate conserved elements are embedded within broader, weakly constrained transcript contexts, emphasizing the importance of localized comparison in cross-species ncRNA analysis.

Figure 6. Modular and heterogeneous cross-species correspondence of lncRNA cores.
(A) Cross-species matching of core regions for representative lncRNAs (NORAD, PVT1, FIRRE). Rows correspond to species and columns to reference cores ordered along the transcript; color indicates MMD, and gray denotes absence of a supported match. Core positions along the reference transcript are shown below each heatmap.
(B) Distribution of core reproducibility across species, shown as the fraction of cores detected in ≥k species.
(C) Median island-level MMD across species (± IQR), showing a consistent increase with phylogenetic distance.

4.4 Limitations

This study has several limitations related to data availability, modeling assumptions, and evaluation. First, the lack of reliable ground truth for cross-species ncRNA correspondence remains a major challenge. Existing annotations are incomplete and heterogeneous across species, and transcript boundaries—particularly for lncRNAs—are often ambiguous. As a result, evaluation based on annotation overlap provides only an approximate and conservative estimate of performance.
Second, CURIA depends on the availability and quality of genome alignment chains. While chains effectively constrain the search space, errors or fragmentation in alignments can affect downstream predictions. In addition, the candidate-region classification step is adapted from protein-coding gene projection and may not fully capture the diversity of ncRNA correspondence patterns.
Third, the method relies on embedding representations derived from RNA foundation models. Although RNA-FM provides a practical representation, it may not capture all relevant structural features. In particular, CURIA is primarily sensitive to localized signal in embedding space and does not model long-range interactions or global RNA architecture.

Fourth, CURIA does not establish transcriptional activity or molecular function. Predicted regions represent candidate corresponding elements and require additional transcriptomic or experimental validation.
Fifth, absence of overlap with annotated loci does not necessarily indicate false positives, and reported performance likely underestimates true recall due to annotation incompleteness.
Finally, CURIA does not explicitly distinguish between shared evolutionary origin and convergent similarity at the level of embedding space. While synteny strongly biases predictions toward corresponding loci with shared ancestry, embedding similarity may also capture convergent structural features or recurrent local motifs. Accordingly, high-confidence matches should be interpreted as candidate corresponding elements supported by both synteny and embedding similarity, rather than definitive evidence of shared evolutionary origin.

4.5 Embedding-based similarity and interpretability

Embedding representations used in CURIA are derived from RNA foundation models trained on large collections of RNA sequences without explicit structural supervision[16,27,28]. Prior work has shown that such models encode representations associated with RNA secondary structure and related functional properties, and can support downstream structure prediction tasks[37].
In this context, embedding similarity provides an implicit measure of correspondence in embedding space that is not tied directly to primary-sequence identity. Empirically, MMD values form a continuous scale separating high-confidence matches from background signal: low values are associated with stronger annotation agreement and more reproducible cross-species correspondence, whereas higher values correspond to weak or ambiguous matches. This interpretation is broadly consistent with observations from protein language models, where representations learned from sequence alone organize according to structural and functional properties[38].
Overall, embedding-based similarity provides a continuous empirical measure of correspondence confidence without requiring explicit structural alignment, although it does not directly model RNA structure or function.

4.6 Failure modes

The limitations of CURIA follow directly from its design assumptions.
First, the method depends on the presence of localized signal in the embedding space. Loci lacking such signal are unlikely to yield detectable correspondence.
Second, CURIA relies on genome alignment chains to constrain the search space. As a result, correspondence cannot be recovered outside aligned regions, and errors in alignments propagate directly to predictions.
Third, the approach assumes approximate collinearity and focuses on local similarity. It therefore does not capture rearrangements of structured elements or long-range RNA interactions.
These constraints imply that CURIA is most effective for loci with well-localized embedding-supported signal and reliable synteny, and that absence of a detected match does not necessarily indicate absence of biological correspondence.

4.7 Future directions

Several directions follow naturally from the current design and limitations of CURIA.

4.7.1 Improved representations for RNA structure and context

CURIA relies on embedding representations that are local and context-dependent. This enables fine-scale matching, but also introduces sensitivity to boundary definition and limits the ability to capture broader structural context. An important next step is the development of representations that remain stable across varying sequence boundaries while preserving the ability to localize signal within longer regions. Such representations could support a multi-resolution framework in which candidate regions are first identified at coarse scale and then refined locally. This may require models trained explicitly for structural consistency across contexts, or alternative similarity measures beyond MMD that better capture structured relationships between regions.

4.7.2 Refinement of synteny-based constraints and evolutionary interpretation

In the current formulation, genome alignment chains are used primarily to restrict the search space. Future work could make more flexible use of genomic context, including weaker forms of syntenic evidence such as conserved neighborhood structure, intronic or antisense relationships, and proximity to orthologous protein-coding genes. This would allow candidate regions to be prioritized even when direct chain projection is fragmented or absent.
A related challenge is the interpretation of matched regions in evolutionary terms. Embedding similarity can identify candidate correspondences, but does not by itself establish orthology. Incorporating additional evidence such as multi-species consistency, conservation of local genomic context, and phylogenetic patterns could help distinguish shared evolutionary origin from convergent structural similarity. These extensions would move the framework from correspondence detection toward a more explicit treatment of ncRNA orthology.

4.7.3 Multi-species integration and consistency

CURIA currently operates in a pairwise setting, whereas ncRNA correspondence is inherently a multi-species problem. Extending the framework to jointly model multiple species would enable the use of phylogenetically informed consistency constraints, improving robustness and reducing false positives. In particular, recurrent low-MMD matches across species could be incorporated directly into the inference procedure rather than evaluated post hoc. This would provide a more principled basis for identifying conserved cores and for separating stable evolutionary signal from species-specific or spurious matches.

4.7.4 Toward annotation frameworks

Finally, extending CURIA from a proof-of-concept pipeline to a broader annotation framework will require improvements in sensitivity, robustness, and interpretability. These include better treatment of ambiguous loci, stronger filtering of false positives, clearer confidence calibration, and tighter integration with existing genome annotation ecosystems. The current results suggest that synteny-constrained, embedding-based methods can provide a useful foundation for such systems, but substantial work remains before they can support routine large-scale annotation.

5. Data and code availability

The CURIA pipeline is available at the project repository:
https://github.com/kirilenkobm/CURIA_pipeline
Preprocessed reference annotation files used in this study are provided in the repository under input_data/reference_annotation/. All predicted annotations, intermediate outputs, and summary result files used in the analyses are available under preprint_results/, including the set of candidate conserved intergenic loci described in Section 3.5. Genome assemblies (2bit format) were obtained from NCBI[39] and UCSC Genome Browser[40] resources. Genome alignment chains were obtained from the TOGA project (Hiller lab)[1]: https://genome.senckenberg.de/download/TOGA. All external datasets are publicly available, and links to the exact sources are provided in the repository.

References

1. Kirilenko BM, Munegowda C, Osipova E, Jebb D, Sharma V, Blumer M, et al. Integrating gene annotation with orthology inference at scale. Science. 2023 Apr 28;380(6643):eabn3107. doi:10.1126/science.abn3107

2. Sharma V, Hiller M. Increased alignment sensitivity improves the usage of genome alignments for comparative gene annotation. Nucleic Acids Res. 2017 Aug 21;45(14):8369–77. doi:10.1093/nar/gkx554

3. Kent WJ, Baertsch R, Hinrichs A, Miller W, Haussler D. Evolution's cauldron: Duplication, deletion, and rearrangement in the mouse and human genomes. Proc Natl Acad Sci. 2003 Sep 30;100(20):11484–9. doi:10.1073/pnas.1932072100

4. Sharma V, Schwede P, Hiller M. CESAR 2.0 substantially improves speed and accuracy of comparative gene annotation. Kelso J, editor. Bioinformatics. 2017 Dec 15;33(24):3985–7. doi:10.1093/bioinformatics/btx527

5. Herrero J, Muffato M, Beal K, Fitzgerald S, Gordon L, Pignatelli M, et al. Ensembl comparative genomics resources. Database. 2016;2016:bav096. doi:10.1093/database/bav096

6. Fitch WM. Distinguishing Homologous from Analogous Proteins. Syst Zool. 1970 Jun;19(2):99. doi:10.2307/2412448

7. Koonin EV. Orthologs, Paralogs, and Evolutionary Genomics. Annu Rev Genet. 2005 Dec 1;39(1):309–38. doi:10.1146/annurev.genet.39.073003.114725

8. Ulitsky I, Bartel DP. lincRNAs: Genomics, Evolution, and Mechanisms. Cell. 2013 Jul;154(1):26–46. doi:10.1016/j.cell.2013.06.020

9. Diederichs S. The four dimensions of noncoding RNA conservation. Trends Genet. 2014 Apr;30(4):121–3. doi:10.1016/j.tig.2014.01.004

10. Hezroni H, Koppstein D, Schwartz MG, Avrutin A, Bartel DP, Ulitsky I. Principles of Long Noncoding RNA Evolution Derived from Direct Comparison of Transcriptomes in 17 Species. Cell Rep. 2015 May;11(7):1110–22. doi:10.1016/j.celrep.2015.04.023

11. Rivas E. Evolutionary conservation of RNA sequence and structure. WIREs RNA. 2021 Sep;12(5):e1649. doi:10.1002/wrna.1649

12. Ulitsky I. Evolution to the rescue: using comparative genomics to understand long non-coding RNAs. Nat Rev Genet. 2016 Oct;17(10):601–14. doi:10.1038/nrg.2016.85

13. Statello L, Guo CJ, Chen LL, Huarte M. Gene regulation by long non-coding RNAs and its biological functions. Nat Rev Mol Cell Biol. 2021 Feb;22(2):96–118. doi:10.1038/s41580-020-00315-9

14. Mattick JS, Amaral PP, Carninci P, Carpenter S, Chang HY, Chen LL, et al. Long non-coding RNAs: definitions, functions, challenges and recommendations. Nat Rev Mol Cell Biol. 2023 Jun;24(6):430–47. doi:10.1038/s41580-022-00566-8

15. Pertea M, Shumate A, Pertea G, Varabyou A, Breitwieser FP, Chang YC, et al. CHESS: a new human gene catalog curated from thousands of large-scale RNA sequencing experiments reveals extensive transcriptional noise. Genome Biol. 2018 Dec;19(1):208. doi:10.1186/s13059-018-1590-2

16. Chen J, Hu Z, Sun S, Tan Q, Wang Y, Yu Q, et al. Interpretable RNA Foundation Model from Unannotated Data for Highly Accurate RNA Structure and Function Predictions [Internet]. arXiv; 2022 [cited 2026 Mar 12]. Available from: http://arxiv.org/abs/2204.00300 doi:10.48550/arXiv.2204.00300

17. Nawrocki EP, Eddy SR. Infernal 1.1: 100-fold faster RNA homology searches. Bioinformatics. 2013 Nov 15;29(22):2933–5. doi:10.1093/bioinformatics/btt509

18. Ontiveros-Palacios N, Cooke E, Nawrocki EP, Triebel S, Marz M, Rivas E, et al. Rfam 15: RNA families database in 2025. Nucleic Acids Res. 2025 Jan 6;53(D1):D258–67. doi:10.1093/nar/gkae1023

19. Will S. LocARNA 2.0: Versatile Simultaneous Alignment and Folding of RNAs. In: Lorenz R, editor. RNA Folding [Internet]. New York, NY: Springer US; 2024 [cited 2026 Mar 22]. p. 235–54. (Methods in Molecular Biology). Available from: https://link.springer.com/10.1007/978-1-0716-3519-3_10 doi:10.1007/978-1-0716-3519-3_10

20. Höchsmann M, Töller T, Giegerich R, Kurtz S. Local similarity in RNA secondary structures. Proc IEEE Comput Soc Bioinforma Conf. 2003;2:159–68. PubMed PMID: 16452790.

21. Gruber AR, Findeiß S, Washietl S, Hofacker IL, Stadler PF. RNAz 2.0: improved noncoding RNA detection. Pac Symp Biocomput Pac Symp Biocomput. 2010;69–79. PubMed PMID: 19908359.

22. Pedersen JS, Bejerano G, Siepel A, Rosenbloom K, Lindblad-Toh K, Lander ES, et al. Identification and Classification of Conserved RNA Secondary Structures in the Human Genome. Durbin R, editor. PLoS Comput Biol. 2006 Apr 21;2(4):e33. doi:10.1371/journal.pcbi.0020033

23. Wucher V, Legeai F, Hédan B, Rizk G, Lagoutte L, Leeb T, et al. FEELnc: a tool for long non-coding RNA annotation and its application to the dog transcriptome. Nucleic Acids Res. 2017 Jan 3;gkw1306. doi:10.1093/nar/gkw1306

24. Zoonomia Consortium, Genereux DP, Serres A, Armstrong J, Johnson J, Marinescu VD, et al. A comparative genomics multitool for scientific discovery and conservation. Nature. 2020 Nov 12;587(7833):240–5. doi:10.1038/s41586-020-2876-6

25. Armstrong J, Hickey G, Diekhans M, Fiddes IT, Novak AM, Deran A, et al. Progressive Cactus is a multiple-genome aligner for the thousand-genome era. Nature. 2020 Nov 12;587(7833):246–51. doi:10.1038/s41586-020-2871-y

26. Penić RJ, Vlašić T, Huber RG, Wan Y, Šikić M. RiNALMo: general-purpose RNA language models can generalize well on structure prediction tasks. Nat Commun. 2025 Jul 1;16(1):5671. doi:10.1038/s41467-025-60872-5

27. Akiyama M, Sakakibara Y. Informative RNA base embedding for RNA structural alignment and clustering by deep representation learning. NAR Genomics Bioinforma. 2022 Jan 13;4(1):lqac012. doi:10.1093/nargab/lqac012

28. Vahab N, Tyagi S. Advancing non-coding RNA annotation with RNA sequence foundation models: structure and function perspectives. BMC Artif Intell. 2025 Oct 14;1(1):12. doi:10.1186/s44398-025-00012-7

29. Gretton A, Borgwardt KM, Rasch MJ, Schölkopf B, Smola A. A Kernel Two-Sample Test. J Mach Learn Res. 2012;13(25):723–73.

30. Smith TF, Waterman MS. Identification of common molecular subsequences. J Mol Biol. 1981 Mar;147(1):195–7. doi:10.1016/0022-2836(81)90087-5

31. Harrow J, Frankish A, Gonzalez JM, Tapanari E, Diekhans M, Kokocinski F, et al. GENCODE: The reference human genome annotation for The ENCODE Project. Genome Res. 2012 Sep;22(9):1760–74. doi:10.1101/gr.135350.111

32. Chan PP, Lin BY, Mak AJ, Lowe TM. tRNAscan-SE 2.0: improved detection and functional classification of transfer RNA genes. Nucleic Acids Res. 2021 Sep 20;49(16):9077–96. doi:10.1093/nar/gkab688

33. Quinn JJ, Chang HY. Unique features of long non-coding RNA biogenesis and function. Nat Rev Genet. 2016 Jan;17(1):47–62. doi:10.1038/nrg.2015.10

34. Wilusz JE, JnBaptiste CK, Lu LY, Kuhn CD, Joshua-Tor L, Sharp PA. A triple helix stabilizes the 3′ ends of long noncoding RNAs that lack poly(A) tails. Genes Dev. 2012 Nov 1;26(21):2392–407. doi:10.1101/gad.204438.112

35. Yamazaki T, Souquere S, Chujo T, Kobelke S, Chong YS, Fox AH, et al. Functional Domains of NEAT1 Architectural lncRNA Induce Paraspeckle Assembly through Phase Separation. Mol Cell. 2018 Jun;70(6):1038-1053.e7. doi:10.1016/j.molcel.2018.05.019

36. Engreitz JM, Pandya-Jones A, McDonel P, Shishkin A, Sirokman K, Surka C, et al. The Xist lncRNA Exploits Three-Dimensional Genome Architecture to Spread Across the X Chromosome. Science. 2013 Aug 16;341(6147):1237973. doi:10.1126/science.1237973

37. Shen T, Hu Z, Sun S, Liu D, Wong F, Wang J, et al. Accurate RNA 3D structure prediction using a language model-based deep learning approach. Nat Methods. 2024 Dec;21(12):2287–98. doi:10.1038/s41592-024-02487-0

38. Rives A, Meier J, Sercu T, Goyal S, Lin Z, Liu J, et al. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. Proc Natl Acad Sci. 2021 Apr 13;118(15):e2016239118. doi:10.1073/pnas.2016239118

39. Sayers EW, Bolton EE, Brister JR, Canese K, Chan J, Comeau DC, et al. Database resources of the national center for biotechnology information. Nucleic Acids Res. 2022 Jan 7;50(D1):D20–6. doi:10.1093/nar/gkab1112

40. Kent WJ, Sugnet CW, Furey TS, Roskin KM, Pringle TH, Zahler AM, et al. The Human Genome Browser at UCSC. Genome Res. 2002 Jun 1;12(6):996–1006. doi:10.1101/gr.229102